

RESEARCH REPORT

Community-Based Breast Cancer Prevention and Early Detection in California, 2014-2023

A decade evidence synthesis of screening access, risk reduction, navigation, and system-level intervention priorities

Evidence window January 2014 through December 2023; current screening guidance used for interpretation

Central conclusion

California's strongest breast cancer strategy is not a single campaign or screening event. It is a connected prevention and early-detection system that combines risk reduction, routine mammography, culturally and linguistically responsive navigation, and reliable diagnostic follow-up.

Executive summary

Breast cancer control in California is often framed as a mammography problem. The evidence reviewed for this report supports a broader conclusion. Mammography is essential secondary prevention, but it does not reduce the incidence of breast cancer and it does not produce benefit unless abnormal findings are resolved through diagnostic imaging, biopsy when indicated, and timely entry into treatment. California's community intervention agenda therefore has to operate across three linked domains: primary prevention, screening access, and completion of the diagnostic pathway [1,2].

The California-focused prevention plan described by White and colleagues reported that the state accounted for 10.8% of U.S. breast cancer cases in 2016 and that mammography use had remained near 72% for roughly a decade. Those figures capture the central policy problem: a high statewide average can coexist with persistent gaps by income, language, geography, insurance, and race or ethnicity [1].

Evidence from the Community Guide, patient-navigation research, and federal safety-net programs consistently favors multicomponent approaches. Interventions work better when reminders and education are paired with barrier reduction, scheduling assistance, transportation support, language-concordant outreach, and navigation after an abnormal result [3-7]. The operational unit is the care cascade, not the outreach contact.

Priority for the next investment cycle

Measure community breast cancer programs by completed care pathways: eligible population reached, screening completed, abnormal findings resolved, diagnosis established, and treatment initiated. Screening volume alone is an incomplete performance measure.

Key findings

- California carried a substantial share of the national breast cancer burden: 10.8% of U.S. breast cancer cases in 2016 were reported in the state [1].
- Statewide mammography use was approximately 72% and had changed little over the preceding decade, indicating that further progress required targeted delivery strategies rather than generic awareness alone [1].
- The 2024 USPSTF recommendation shifted routine biennial mammography to ages 40-74, increasing the size of the population that community programs must identify and support [2].
- Published intervention reviews support multicomponent programs that combine demand generation with access enhancement and provider/system changes [4,5].
- Navigation is most consequential when it continues beyond the mammogram into diagnostic resolution and treatment linkage [6,7].
- Primary prevention deserves a larger role. Breast cancer incidence cannot be reduced through screening alone; community action must also address alcohol exposure, physical inactivity, obesity after menopause, environmental exposures, and access to risk assessment and preventive therapy [1].

1. Background

Breast cancer remains one of the most consequential cancers for women in California, but the burden is not evenly distributed. Population averages can obscure communities in which screening is difficult to obtain, diagnostic facilities are geographically distant, language access is inconsistent, or the cost of missed work and transportation makes a “free” mammogram practically inaccessible. A community intervention must therefore be judged not only by whether it communicates a recommendation, but by whether it changes the conditions that determine completion.

White and colleagues developed a California breast cancer primary-prevention agenda by integrating scientific evidence with community perspectives. Their framework is important because it distinguishes primary prevention

from early detection. Mammography identifies disease earlier; it does not prevent the disease from occurring. A comprehensive state strategy needs both: community environments that reduce avoidable risk and a screening-and-diagnostic system that detects disease at a treatable stage [1].

The clinical context changed near the end of the review period. In April 2024, the USPSTF recommended biennial mammography for women ages 40 through 74. The recommendation was partly responsive to increasing incidence among women in their forties and to the disproportionate mortality experienced by Black women. Community programs designed during 2014-2023 now have to adapt their registries, outreach criteria, referral capacity, and funding to this larger eligible population [2].

2. Methods

This report used a targeted narrative evidence-synthesis design. The primary evidence window was January 2014 through December 2023. Current federal guidance published in 2024 and current federal program documentation were included when needed to interpret implementation priorities after the review period.

Sources were selected from peer-reviewed publications, federal evidence reviews, the California Comprehensive Cancer Control Plan, and government program documentation. Eligible evidence addressed at least one of the following: community breast cancer risk reduction; mammography access; patient navigation; community health workers or promotoras; mobile or decentralized screening; diagnostic follow-up; survivorship support; or implementation outcomes relevant to underserved California populations.

Reported numerical values were extracted as presented by the source. This report did not pool heterogeneous studies, generate county estimates, or infer causal effects from descriptive program data. National values are used as benchmarks and are labeled as national evidence.

Table 1. Source-reported indicators used in this report

Indicator	Reported value	Geographic scope	Interpretation
Share of U.S. breast cancer cases occurring in California, 2016	10.8%	California	Establishes the scale of the state burden [1].
Mammography use reported in the California prevention agenda	Approximately 72%; little change over about a decade	California	A high average can conceal subgroup and geographic gaps [1].
Recommended routine screening interval	Biennial mammography, ages 40-74	United States guideline	Expands community-program eligibility beginning at age 40 [2].
Women served by NBCCEDP since 1991	More than 6.6 million	United States safety-net program	Shows the reach possible through organized public programs [3].
Breast and cervical screening examinations through NBCCEDP	More than 17.0 million	United States safety-net program	Demonstrates longitudinal program scale [3].
Share of estimated breast-screening-eligible women served by NBCCEDP, 2020-2021	13.5%	United States safety-net program	Shows that safety-net demand exceeds program reach [3].

Values are reproduced from the cited publications and program documentation. NBCCEDP = National Breast and Cervical Cancer Early Detection Program.

3. Findings

3.1 Community interventions must address the full prevention continuum

The reviewed evidence separates breast cancer control into three related but distinct functions. Primary prevention seeks to reduce incidence by changing exposures and risk-related environments. Secondary prevention uses mammography and risk-appropriate imaging to identify disease earlier. Tertiary and survivorship interventions

reduce the burden after diagnosis through navigation, symptom management, treatment support, and long-term follow-up. Programs become less effective when these functions are treated as independent projects with separate referral systems and temporary funding.

For primary prevention, the California agenda emphasized multilevel intervention rather than individual education alone. Community-level action can support physical activity, healthy weight, reduced alcohol exposure, breastfeeding, safer occupational and environmental conditions, and equitable access to risk assessment. For people at elevated risk, prevention also includes counseling about risk-reducing medications and genetic services when clinically indicated [1].

3.2 Screening gains are strongest when interventions are multicomponent

Systematic reviews conducted for the Guide to Community Preventive Services found that breast cancer screening can be increased through client reminders, small media, one-on-one education, reduction of structural barriers, and provider-oriented interventions. The strongest implementation logic is additive: use outreach to identify eligible people, remove the barriers that make the appointment difficult, and modify clinic systems so that missed opportunities are visible and acted upon [4,5].

In California, this translates into registries that accurately identify who is due, outreach in the patient's preferred language, scheduling during the outreach encounter, evening or weekend capacity, transportation support, mobile or co-located imaging where access is sparse, and a closed-loop process for abnormal findings. A reminder that leaves the patient to solve every subsequent barrier is not equivalent to navigation.

3.3 Screening is only the first measurable step

The USPSTF explicitly notes that equitable benefit requires follow-up evaluation and treatment after an abnormal mammogram. This distinction is central to community programs because the groups most likely to face barriers to screening often face even greater barriers after a result becomes abnormal. Diagnostic imaging may be offered at a different facility, biopsy may require another authorization, and treatment linkage may involve a new health system. Each transfer creates a new point of loss [2].

Programs should therefore retain responsibility until diagnostic resolution rather than defining success at mammography completion. At minimum, community program dashboards should report abnormal-result notification, diagnostic imaging completion, biopsy completion when recommended, time to definitive diagnosis, and treatment initiation.

3.4 Equity requires targeted capacity, not uniform outreach

National evidence illustrates why equal outreach volume is not enough. The USPSTF evidence review reported breast cancer mortality of 27.6 per 100,000 among non-Hispanic Black women compared with 19.7 per 100,000 among non-Hispanic White women. It also reported triple-negative breast cancer incidence of 24.2 versus 12.3 per 100,000, respectively [2]. These differences reflect both disease biology and unequal access to timely, high-quality care. California programs should use local registry and health-system data to determine where the cascade fails rather than assuming that a statewide screening average represents equitable performance.

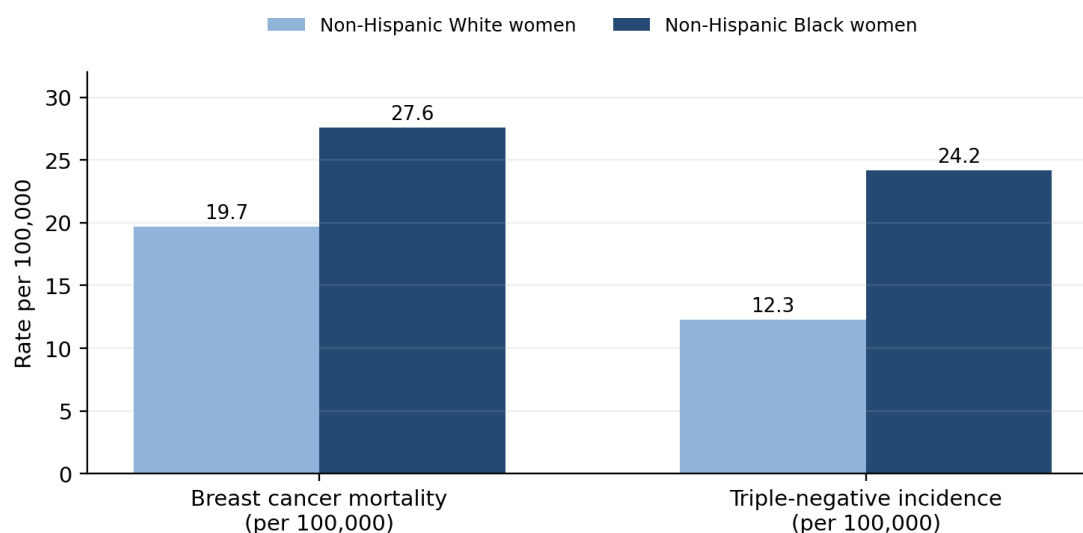


Figure 1. National racial equity benchmarks relevant to California breast cancer program design

Source: U.S. Preventive Services Task Force evidence summarized in the 2024 breast cancer screening recommendation [2]. Values are national, not California-specific.

Table 2. Intervention components and the outcomes they are designed to change

Component	Primary mechanism	Minimum outcome measure	Implementation risk
Population registry and reminders	Identifies people due for screening and reduces missed opportunities	Proportion of eligible population contacted and screened	Incomplete demographic or insurance data can misclassify eligibility.
Promotora/CHW outreach	Builds trust, provides language-concordant education, and identifies barriers	Contact-to-scheduling conversion; screening completion	Education-only roles may not resolve logistical barriers.
Patient navigation	Coordinates scheduling, transportation, authorizations, and follow-up	Diagnostic resolution and time to resolution	Short-term funding can end support before the pathway is complete.
Mobile or co-located mammography	Reduces travel and appointment burden	Completed mammograms among people with access barriers	Without referral agreements, abnormal results may be disconnected from diagnostics.
Provider reminders and audit/feedback	Makes overdue screening visible in clinical workflows	Orders placed; screening completed; missed-opportunity rate	Alert fatigue and lack of imaging capacity can limit effect.
Risk assessment and prevention counseling	Identifies elevated risk and supports prevention choices	Risk assessment completion; referral and preventive therapy uptake	Risk tools and genetic services may perform or reach populations unevenly.

The table describes the intended pathway of action. It is not a pooled estimate of intervention effect.

4. Implications for California communities

The reviewed evidence supports a regionalized strategy. California’s urban, agricultural, frontier, tribal, and border communities do not experience the same barriers, and the delivery model should reflect those differences. Mobile mammography may be essential where distance is the main barrier; extended hours and paid time constraints may dominate in urban working populations; language-concordant navigation and immigration-related trust may be central in other communities. The common requirement is a pathway that is easy to enter and difficult to fall out of.

The state also needs to protect the infrastructure that is least visible in a screening count. Navigation staff, referral coordinators, interpreters, and data personnel are recurring operating costs. When they are funded as temporary outreach enhancements, program capacity disappears when the grant ends even though the underlying need remains.

5. Recommendations

1. Adopt a statewide breast cancer intervention framework that treats primary prevention, screening, diagnostic resolution, and treatment linkage as one continuum.
2. Update community eligibility registries and capacity plans for routine biennial mammography beginning at age 40, while preserving risk-based pathways for people who need earlier or more intensive surveillance.
3. Fund navigation and community health worker positions as ongoing service infrastructure rather than time-limited campaign personnel.
4. Require funded programs to report cascade measures, including time from abnormal mammogram to diagnostic resolution, stratified by race and ethnicity, language, insurance, geography, and disability status.
5. Pair mobile and decentralized screening with formal diagnostic referral agreements and responsibility for closed-loop follow-up.
6. Expand community-facing primary-prevention work, including alcohol risk communication, physical-activity environments, healthy-weight supports, breastfeeding support, and equitable access to risk assessment and preventive therapy.
7. Use implementation measures such as reach, fidelity, adoption, and sustainability alongside clinical outcomes so that successful models can be reproduced across regions.

6. Limitations

This report is a narrative synthesis rather than a systematic review or meta-analysis. The underlying studies vary in population, intervention content, follow-up, and outcome definition. Some California program data are not reported publicly at a level that permits comparison across counties or demographic groups. National program statistics and national racial-disparity estimates are included as contextual benchmarks and are labeled accordingly. The evidence supports the direction of the recommendations, but local implementation decisions should be informed by current county, health-system, and California Cancer Registry data.

7. Conclusion

A decade of breast cancer intervention evidence points to a clear systems lesson for California. Outreach matters, but outreach alone is not a program. The effective unit is a connected pathway that reduces risk where possible, identifies eligible people, makes mammography accessible, accompanies patients through abnormal findings, and links confirmed disease to treatment. California can improve both population outcomes and equity by financing that pathway as durable public-health infrastructure.

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